Carmine Potirniche

Reno, NV | (725) 910-8625 | potirnichecarmine@gmail.com
carminepotirniche.com | linkedin.com/in/carmineap | github.com/Potirniche-Carmine

Experience

Crytica Security Reno, NV
Software Engineer | *Cybersecurity startup* *Nov. 2025 – Present*

- Architected and built Crytica's multi-tenant administration platform as sole developer: a C backend service, CLI, and JavaScript/PHP web console with **100+ API operations**, from requirements to release.
- Designed its security model: privilege-separated services, a schema-validated binary IPC protocol, database-enforced RBAC with per-tenant scoping, and a fixed catalog of **300+ prepared SQL statements**.
- Built the RHEL 9 installer and setup wizard, rewrote the SSH deployment daemon, and wrote a checksum-verified schema migrator; cut full deployments across **3–12 servers** to **under 5 minutes**.
- Orchestrated **hundreds of AI agents and subagents**, gating their work with code review, **2,600+ unit tests** run under sanitizers in CI, and black-box end-to-end installs of multi-server environments.
- Led a C synchronization rewrite across **9+ distributed ledgers**, cutting sync time by **85%**.

Projects

Umbratic | *AI Workspace & Agent Runtime* Mar. 2025 – Present

- Built an Electron/React AI workspace backed by a Rust agent runtime and Go model gateway, supporting local inference, self-hosted models, and integrations with Codex and Claude Code.
- Implemented an **agent execution harness** with dependency-aware parallel tools, context compaction, resumable state, bounded retries, cancellation, and permission checks for workspace operations.
- Implemented team-filtered Qdrant retrieval, device enrollment, and revocation for managed model access.
- Built tools to create and edit documents, spreadsheets, presentations, and PDFs within the AI workspace.

GamePack | *Native Video Review Application* Oct. 2026 – Present

- Built a cross-platform video review application with React Native/TypeScript, native AVPlayer/MediaPlayer playback, and a Rust/SQLite engine; synchronized frame and timed annotations with threaded feedback.
- Designed durable local storage with transactional comment posting, recoverable drafts, and BLAKE3-verified media imports to prevent duplicate copies and make retries idempotent.
- Automated CI/CD for **macOS and Windows on ARM64/x64**, including native tests, macOS signing and notarization, SHA-256-verified updates, and rollback on failed installation.

UpDrafted | *Athlete Recruiting Platform* May 2025 – May 2026

- Built Next.js/TypeScript recruiting and administration apps with athlete discovery, verification, moderation, Stripe subscriptions, signed webhooks, and Cloudflare R2 storage.
- Implemented **14 PostgreSQL row-level security policies**, transaction-scoped user context, and AES-256-GCM message encryption; automated Docker builds and deployments with GitHub Actions.

Education

University of Nevada, Reno Reno, NV
B.S. in Computer Science and Engineering, Minor in Statistics *May 2025*

Technical Skills

Languages: C, TypeScript/JavaScript, Rust, Go, Python, SQL, PHP, Bash, C#
Systems & Delivery: Linux (RHEL), systemd, SELinux, NGINX, CMake, Docker, GitHub Actions, CI/CD
Frameworks & Data: React, React Native, Next.js, Electron, MariaDB/MySQL, PostgreSQL, REST APIs
Security & Testing: RBAC, row-level security, privilege separation, encryption, ASan/UBSan, Playwright
AI & Agents: Agent orchestration, tool calling, context compaction, RAG, Qdrant, llama.cpp, vLLM